

## Acknowledgments.

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## A Why These Three Designs?

The choice of binary, exponential, and PageRank rewards is motivated by three fundamental principles.

**Minimality and Baseline.** The binary reward represents the minimal verifiable reward: “does the completion contain any forbidden item?” This makes it the natural baseline. It is also the instantiation implicitly used in PURGE [37], establishing a clear comparison point. Any improvement beyond binary represents a genuine gain from richer information extraction.

**Smooth Relaxation along Information Hierarchy.** The exponential reward sits directly between binary and PageRank on the information hierarchy. It uses count-level information ( $I_{\text{count}}$ ) rather than binary presence, but makes no structural assumptions about the forget set. This makes it a natural intermediate step that isolates the effect of moving from discrete to continuous reward signals.

**Exploitation of Natural Structure.** The PageRank reward exploits the hierarchical structure that naturally exists in extracted forget sets. PURGE and related work extract entities via NER, producing a semantic hierarchy: the queried entity is primary, and associated concepts (attributes, related items) are secondary. PageRank formalizes this intuition by computing importance weights directly from embeddings and semantic similarity. It represents the richest information component that remains verifiable without access to training data.

*Remark 1.* Extensions beyond PageRank (e.g., learned information components, adaptive weights) would require additional modeling assumptions or data beyond what is verifiable. Thus, PageRank represents a natural stopping point for the verifiable reward hierarchy.

## B Convergence Analysis: Why Denser Rewards are Faster

To make precise the claim that PageRank  $\succ$  Exponential  $\succ$  Binary in convergence speed, we analyze how reward richness affects the learning signal.

### B.1 Advantage Variance under GRPO

Recall that GRPO computes token-level advantages as:

$$\hat{A}_\phi(o) = \frac{\phi(o) - \mu}{\sigma} \quad (14)$$

where  $\mu = \mathbb{E}_{o \sim \text{batch}}[\phi(o)]$  and  $\sigma = \text{std}_{o \sim \text{batch}}[\phi(o)]$  are sample statistics from a batch of completions  $\{o_1, \dots, o_G\}$ .

**Lemma 1 (Reward Range and Advantage Variance).** *For any reward function  $\phi : \mathcal{Y} \rightarrow [0, 1]$ , the variance of advantage estimates satisfies:*

$$\text{Var}[\hat{A}_\phi] \leq \frac{\text{Var}[\phi]}{\sigma^2}. \quad (15)$$

*The denominator  $\sigma^2$  depends on the reward distribution. If rewards cluster at the boundary (many zeros, few ones),  $\sigma \rightarrow 0$ , and advantages become unstable. If rewards are spread across  $[0, 1]$ ,  $\sigma$  remains bounded away from zero, keeping advantages stable.*

## B.2 Analysis of Binary Reward

For the binary reward, the support is  $\{0, 1\}$ . Let  $p$  be the probability that a sampled completion has zero forbidden items:

$$p = \mathbb{P} \left[ \sum_j m_j(y) = 0 \right] \quad (16)$$

The empirical reward distribution from a batch of size  $G$  is approximately:

$$\phi(o) \in \{0, 1\}, \quad \mathbb{E}[\phi] \approx p, \quad \text{Var}[\phi] \approx p(1 - p) \quad (17)$$

Therefore:

$$\sigma_{\text{bin}} = \sqrt{p(1 - p)} \quad (18)$$

*Claim (Advantage Instability at Boundaries).* When  $p \rightarrow 0$  or  $p \rightarrow 1$ , we have  $\sigma_{\text{bin}} \rightarrow 0$ . This causes advantage estimates to collapse, losing the learning signal precisely when guidance is needed (early training) or when performance is saturating.

## B.3 Analysis of Exponential Reward

For the exponential reward, the reward takes values in  $(0, 1]$  continuously. Let  $C = c(y; F)$  follow some empirical distribution with mean  $\mu_C$  and variance  $\sigma_C^2$ . By Taylor expansion around  $\mu_C$ :

$$\text{Var}[\phi_{\text{exp}}] \approx \frac{1}{\tau^2} \cdot \text{Var}[C] = \frac{\sigma_C^2}{\tau^2} \quad (19)$$

The empirical standard deviation of rewards is:

$$\sigma_{\text{exp}} \approx \frac{\sigma_C}{\tau} \cdot \phi_{\text{exp}}(\mu_C) \quad (20)$$

*Claim (Maintained Variance Across Training).* Unlike the binary case,  $\sigma_{\text{exp}}$  depends on the distribution of violation counts, not on a boundary probability. Even when most completions have zero violations ( $\mu_C \approx 0$ ), the exponential reward maintains  $\phi_{\text{exp}} \approx 1$  with gradient  $-1/\tau$ . Completions with  $c = 1$  violation yield  $\phi_{\text{exp}} \approx e^{-1/\tau} < 1$ , preserving advantage signals. The variance  $\text{Var}[\phi_{\text{exp}}]$  remains nonzero as long as completion quality varies.

#### B.4 Analysis of PageRank Reward

The PageRank reward takes continuous values in  $[0, 1]$ . Let  $P(y; F) = \sum_{j=1}^m w_j \cdot \mathbf{1}[m_j(y) > 0]$  denote the total weighted penalty.

$$\phi_{\text{pr}} = \text{clip}(1 - P, 0, 1) \quad (21)$$

The key difference from exponential is that PageRank weights penalties by semantic importance. Violations of high-importance items contribute more to  $P$  than violations of peripheral items.

*Claim (Finer-Grained Advantage Signals).* For two completions with the same total violation count  $c(y_1) = c(y_2) = 2$ :

- If  $y_1$  violates two core items:  $P(y_1) = 2w_{\text{core}} \approx 2 \cdot 0.9 = 1.8$ ,  $\phi_{\text{pr}}(y_1) \approx 0$ ;
- If  $y_2$  violates two peripheral items:  $P(y_2) = 2w_{\text{periph}} \approx 2 \cdot 0.1 = 0.2$ ,  $\phi_{\text{pr}}(y_2) \approx 0.8$ ;

Exponential would assign both equal low reward (approximately  $e^{-2/\tau} \approx 0.14$  for moderate  $\tau$ ), failing to distinguish progress. PageRank clearly differentiates, providing stronger learning signals about which violations matter most.

#### B.5 Convergence Rate Comparison

**Theorem 1 (Convergence Speed Ordering).** *Under reasonable assumptions on the violation distribution, the convergence rates satisfy:*

$$\text{rate}_{\text{bin}} < \text{rate}_{\text{exp}} < \text{rate}_{\text{pr}} \quad (22)$$

More formally, the number of training iterations  $T$  required to reach a target unlearning performance  $\epsilon$  scales as:

$$T_{\text{bin}} \propto \frac{1}{p(1-p)} \cdot d(\epsilon) \quad (23)$$

$$T_{\text{exp}} \propto \tau^2 \cdot d(\epsilon) \quad (24)$$

$$T_{\text{pr}} \propto d(\epsilon) \quad (25)$$

where  $d(\epsilon)$  is a problem-dependent term related to forgetting difficulty (forget set size, model capacity), and  $p$  is the success probability under binary rewards.

*Proof (Proof Sketch).* The policy gradient step size is roughly  $\eta \cdot \mathbb{E}[\|\hat{A}\|]$  (ignoring clipping and other details). For gradient magnitude, we have:

$$\mathbb{E}[\|\hat{A}_{\text{bin}}\|] \approx \text{constant} \cdot \sqrt{p(1-p)} \quad (26)$$

$$\mathbb{E}[\|\hat{A}_{\text{exp}}\|] \approx \text{constant} \cdot \frac{\sigma_C}{\tau} \quad (27)$$

$$\mathbb{E}[\|\hat{A}_{\text{pr}}\|] \approx \text{constant} \cdot \sigma_P \quad (28)$$

where  $\sigma_C$  is the std of violation counts and  $\sigma_P$  is the standard deviation of weighted penalties. Crucially:

- (1) **For binary:** As training progresses,  $p \rightarrow 1$  (success rate increases), so  $\sqrt{p(1-p)} \rightarrow 0$ . Gradient collapse occurs near convergence.
- (2) **For exponential:** The std of counts  $\sigma_C$  only goes to zero if all completions converge to identical violation counts. With proper exploration, this remains nonzero. However, the decaying constant  $\tau$  introduces a constant slowdown factor.
- (3) **For PageRank:** The std of weighted penalties  $\sigma_P$  benefits from semantic structure. Even with a few total violations, penalties vary semantically, maintaining the signal. No decaying constant tuning parameter needed.

The result follows from standard optimization theory: more stable gradients with lower variance enable larger step sizes and faster convergence.

## C Experimental Details

### C.1 The RWKU Benchmark

We evaluate all reward variants using the Real-World Knowledge Unlearning (RWKU) benchmark [13], a comprehensive framework designed to assess unlearning in a practical, zero-shot setting where neither the forget corpus nor the retain corpus is provided. RWKU uses 100 real-world famous people as unlearning targets, selected based on Wikipedia popularity rankings. The benchmark is structured around four evaluation splits, each targeting a distinct aspect of the unlearned model.

*Forget Set.* The forget set measures unlearning efficacy across three probe types:

- **Fill-in-the-Blank (FB):** Cloze-style sentences extracted from the target’s Wikipedia page, with key facts masked. Performance is measured via ROUGE-L recall (lower is better ↓).
- **Question-Answering (QA):** Paraphrased and restructured question–answer pairs requiring active knowledge recall. Evaluated with ROUGE-L recall (lower is better ↓).
- **Adversarial Attacks (AA):** Nine jailbreak-style probe types designed to elicit residual knowledge from the unlearned model, including prefix injection, affirmative suffix, role playing, multiple choice, reverse query, synonym manipulation, background hint, in-context learning, and cross-lingual queries. Evaluated with ROUGE-L recall (lower is better ↓).

*Neighbor Set.* The neighbor set evaluates whether unlearning is precise and does not inadvertently suppress knowledge adjacent to the forget target (e.g., facts about a target’s works or associated entities, rather than the target itself). It uses the same FB and QA probe formats as the forget set, but evaluated with ROUGE-L recall (higher is better ↑).

*MIA Set.* The Membership Inference Attack (MIA) set serves as a privacy proxy, assessing whether the model still leaks membership signals for forget-target fragments. It consists of two splits:

- **FM (Forget Members):** Textual fragments drawn from the unlearning target.
- **RM (Retain Members):** Unrelated control fragments.

We primarily report the LOSS-based MIA score. A well-unlearned model should yield higher LOSS on FM relative to RM, indicating weaker membership signals.

*Utility Set.* The utility set quantifies collateral effects on broader model capabilities, using the following benchmarks:

- **General Ability (GA):** MMLU, 5-shot accuracy.
- **Reasoning Ability (RA):** Big-Bench-Hard (BBH), 3-shot chain-of-thought, exact match.
- **Truthfulness (TRU):** TruthfulQA MC1, 6-shot accuracy.
- **Factuality (FAC):** TriviaQA, 6-shot F1.
- **Fluency (FLU):** AlpacaEval, weighted average of bi- and tri-gram entropy (reported as a sum).

All utility metrics are higher-is-better ( $\uparrow$ ). FM and FLU scores are reported as sums over targets, following the RWKU convention.

*Dataset Statistics.* RWKU contains 13,131 multi-level forget probes (3,268 fill-in-the-blank, 2,879 question-answer, 6,984 adversarial attack) and 11,379 neighbor probes. The utility set samples 171 instances from MMLU, 81 from BBH, 50 from TruthfulQA, 100 from TriviaQA, and 50 from AlpacaEval per unlearning target.

## C.2 Hyperparameter Settings

All experiments are conducted using **Phi-3-Mini-4K-Instruct**, a 3.8B-parameter instruction-tuned language model. We use the default PURGE hyperparameters throughout. The group size for sampling is  $G = 8$  completions per query. The PPO clipping threshold is  $\varepsilon = 0.2$ , and the KL penalty weight is  $\beta = 0.001$ . We train for a maximum of 1,500 steps and report results at convergence. All experiments are executed on a system with a single AMD EPYC 7002/3 64-core CPU and one NVIDIA Tesla H200 GPU. We reproduce the performance of the original model (BASE) and the PURGE binary-reward baseline using the default hyperparameters reported in the original PURGE work [37].

*Reward-Specific Hyperparameters.*

- **Binary Reward ( $\phi_{\text{bin}}$ ):** No additional hyperparameters; threshold fixed at zero for forbidden matches.

- **Exponential Reward ( $\phi_{\text{exp}}$ ):** decaying constant parameter  $\tau$ , swept over  $\{0.1, 0.25, 0.5, 1.0, 2.0, 5.0\}$ . Based on ablation results (Section 6.3), we set  $\tau^* = 0.5$  as the default.
- **PageRank Reward ( $\phi_{\text{pr}}$ ):** Semantic graph constructed with cosine-similarity threshold  $\theta = 0.5$ ; personalized PageRank damping factor  $\alpha = 0.85$ . We use the **PageRank-Softmax** variant as the default, based on the ablation in Section 6.3.

## D Additional PageRank Reward Analysis

This appendix provides a comprehensive analysis of PageRank weight redistribution strategies, complementing the ablation study in Section 6.3. We examine how different normalisation schemes affect the weight distribution over the forget set, and present full experimental results across all PageRank variants.

### D.1 Weight Distribution Across Variants

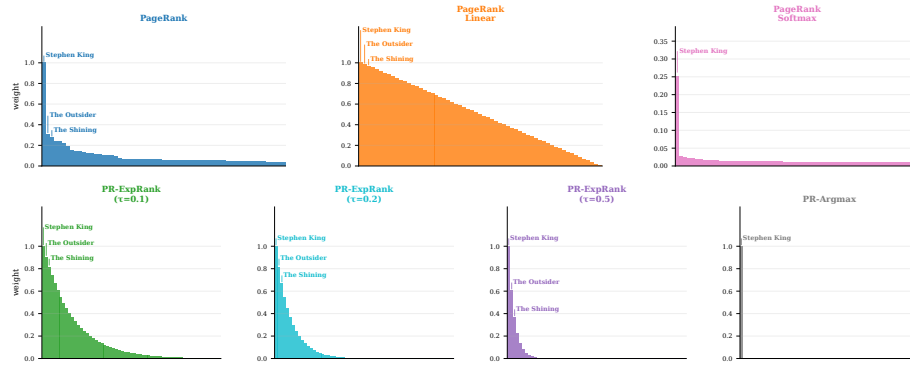
A fundamental challenge with PageRank applied to semantic forget-set graphs is the emergence of a Zipf/power-law weight distribution: a small number of high-centrality nodes (typically the primary forget target and its closest semantic neighbours) capture the vast majority of the total weight mass, leaving peripheral concepts with negligible penalty contribution. This collapse undermines the reward’s ability to provide informative gradients when the model has already suppressed the most salient concepts but still leaks information through secondary entities.

Figure 4 illustrates this effect for the Stephen King forget set (61 terms). PageRank exhibit power-law profile, with the top-5 nodes accounting for 34% of the total weight mass. PageRank-Softmax applies a temperature-scaled softmax over the raw scores, converting them into a proper probability distribution; this compresses the gap between high- and low-weight nodes, while retaining the semantic ordering induced by PageRank. PageRank-Linear discards score magnitudes entirely, replacing them with a linearly spaced sequence from 1 (highest-ranked) to 0 (lowest-ranked); although it corrects the power-law collapse, it does so at the cost of introducing a uniform spacing distortion that is insensitive to the true score gaps between adjacent ranks.

### D.2 Full Experimental Results

Figure 5 reports the complete evaluation of all PageRank variants on the RWKU benchmark after 1,500 training steps, including per-group ranks, win counts, and a robustness score defined as the inverse of the mean standard deviation across forget targets (higher robustness indicates more consistent performance across the 20 unlearning targets).

PageRank-Softmax achieves the best overall rank (0.64) and the highest average improvement over the Base model (+8.1%), with particularly strong performance on Forget metrics (Forget rank 0.95 on FB). Crucially, it also achieves



**Fig. 4.** PageRank weight distribution variants for the Stephen King forget set (61 terms). PageRank exhibit a power-law distribution where the top-5 nodes capture 34% of total weight mass. PageRank-Softmax compresses this gap while preserving semantic ordering; PageRank-Linear corrects the collapse but introduces uniform spacing insensitive to true score gaps. The exprank variants ( $\alpha \in \{0.1, 0.2, 0.5\}$ ) offer a continuous interpolation between these extremes, while PageRank-Argmax concentrates all weight on the single highest-ranked node.

the highest robustness score among the four main variants, suggesting that soft weight redistribution not only improves average-case forgetting but also reduces variance across targets, a practically important property when deploying unlearning at scale.

PageRank-Linear scores worst overall (rank 0.33) and wins no individual metric comparisons, confirming that hard ordinal reassignment introduces a different distortion that ultimately impairs learning.

We also evaluate three exponential-rank variants (PageRank-ExpRank with  $\tau \in \{0.1, 0.2, 0.5\}$ ), which apply an exponential decay to the rank ordering rather than a linear one. These variants interpolate between the hard ordinal signal of PageRank-Linear and the smooth compression of PageRank-Softmax. At  $\tau = 0.5$ , PageRank-ExpRank achieves competitive utility scores (Utility rank 0.77) and strong MIA resistance (MIA rank 1.00), but underperforms PageRank-Softmax on forgetting quality. The PageRank-Argmax variant, which concentrates all weight on the single highest-ranked node, produces the worst forgetting performance (Forget FB 0.467) despite high utility stability, confirming that weight concentration, rather than redistribution, is counterproductive for unlearning.

### D.3 Qualitative Weight Visualisation

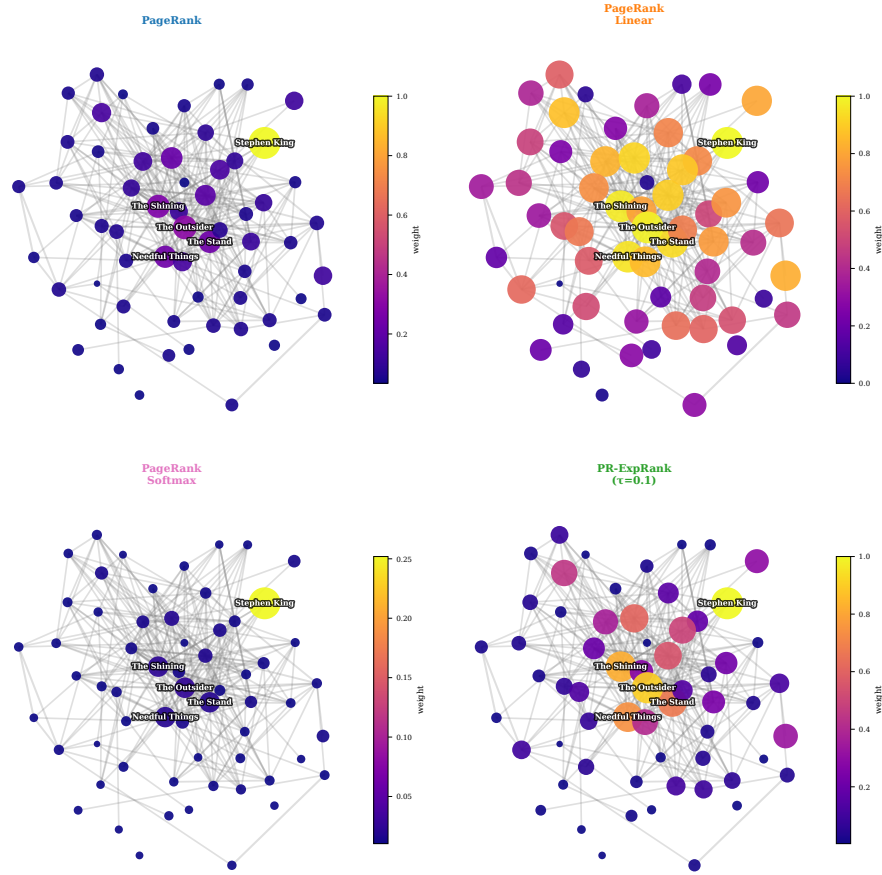
Figure 6 shows the semantic graph and per-node weight distributions for the Stephen King forget set under all four main variants. In the PageRank, the *Stephen King* node dominates visually and numerically, with *The Shining* and *The Stand* forming a high-weight core before the distribution drops sharply.



**Fig. 5.** Evaluation of PageRank variants on the RWKU benchmark at 1,500 training steps, averaged over 20 forget targets. Columns report per-metric scores (Forget, Neighbor, MIA, Utility) alongside per-group ranks, win counts, overall rank, and a robustness score (inverse mean standard deviation across targets; higher = more consistent). PageRank-Softmax achieves the best overall rank and highest robustness, while PageRank-Linear performs worst. PageRank-Argmax and PageRank-ExpRank variants occupy intermediate positions.

PageRank-Softmax visibly redistributes weight across the graph, with more nodes rendered in mid-range colours, while PageRank-Linear produces a near-uniform gradient that treats all 61 nodes as roughly equivalent in importance.

Taken together, these results support the conclusion from Section 6.3: *soft redistribution of weights, rather than hard re-ranking or simple rescaling, is the more effective strategy for PageRank-based reward design*. By compressing rather than discarding score information, PageRank-Softmax retains the semantic ordering induced by graph centrality while spreading penalty mass more evenly across the forget set, resulting in denser and more stable learning signals throughout training.



**Fig. 6.** PageRank graph visualisations for the Stephen King forget set (61 terms) across four weight-transform variants. Node colour and size reflect per-node weight. PageRank distributes penalty mass more evenly than PageRank, while PageRank-Linear yields a near-uniform gradient.